

Projecting harmful algal blooms in North Biscayne Bay using machine learning and bias-corrected climate scenarios

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ABSTRACT

Study area: North Biscayne Bay, downstream of urbanized Miami-Dade County, is an ecological hot spot and frequently impacted by harmful algal blooms (HABs).

Study focus: This study combines machine learning models with bias-corrected climate projections to forecast future chlorophyll-a concentrations, a HAB indicator, in North Biscayne Bay from 2021 to 2100. A Support Vector Machine (SVM) model was developed using historical water quality and climate data, then applied to projections from four global climate models under three Shared Socioeconomic Pathways (SSP126, SSP245, SSP585).

New hydrological insights for the region: Results show a steady increase in chlorophyll-a concentrations in all scenarios with SSP126 showing the largest increase at 0.0045 µg/L per year compared to SSP245 at 0.0007 µg/L per year and SSP585 at 0.0005 µg/L per year. SSP126 predicts the most extreme chlorophyll-a events at 36–82 exceedances followed by SSP245 at 32–65 exceedances and SSP585 at 24–54 exceedances. The three most important positive drivers of chlorophyll-a were precipitation, upstream discharge and minimum temperature while wind speed was negative. Data shows chlorophyll-a concentrations were higher during the wet season than the dry season and grew more during the long-term (2071–2100) than the short-term (2021–2050). These projections assume stationarity in key water quality drivers, which may not fully hold under changing land use or management conditions, but they nonetheless provide valuable guidance for anticipating future trends. These findings highlight the value of using climate-adjusted data and interpretable machine learning to forecast future water quality issues in urban coastal areas, supporting predictive tools that can help prioritize wet-season monitoring and early warning strategies in urban estuaries.

1. Introduction

The prevalence of harmful algal blooms (HABs) presents an escalating ecological crisis, with their toxins and mass proliferation causing profound adverse impacts on aquatic ecosystems, local economies, and public health (Anderson et al., 2021). HAB occurrences are increasingly reported across diverse water bodies, from confined freshwater lakes to expansive estuarine and coastal regions, signifying a threatening environmental challenge (Amorim and Moura, 2021; Dai et al., 2023). HAB dynamics are influenced by multiple factors, according to scientific consensus, including excessive nutrients like nitrogen and phosphorus that lead to

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eutrophication, as well as conducive ecological growth conditions (Glibert, 2020; Heisler et al., 2008; Sinha et al., 2017). The profound repercussions of these blooms necessitate a better understanding of their driving factors to mitigate their negative impacts (Anderson et al., 2021; Glibert and Burford, 2017; Yan et al., 2024b). The absence of definitive and consistent knowledge of HAB formation mechanisms highlights the critical need for enhanced research and monitoring efforts, especially as environmental conditions evolve continuously. Additionally, as global temperatures increase and rainfall patterns change, conditions conducive to the formation of HABs are expected to worsen, further complicating efforts to address this environmental challenge (Gobler, 2020; Griffith and Gobler, 2020; Wells et al., 2020).

Climate change is increasingly recognized as a significant factor influencing HAB dynamics by altering temperature regimes, precipitation patterns, and nutrient loading, thereby influencing bloom distribution characteristics (Glibert, 2020; Mardones et al., 2023; Wells et al., 2015). Existing literature suggests that climate change, by causing warming, acidification, and deoxygenation of coastal waters, is linked to the increased frequency and intensity of HABs (Gobler, 2020; Wells et al., 2020). Furthermore, globally increasing yet regionally varying nutrient loads interact with climate change to affect HAB occurrences, potentially exacerbating the toxicity of some species under unbalanced nutrient conditions (Glibert and Burford, 2017). For example, a meta-analysis paper by Brandenburg et al. (2019) showed that marine harmful algae from temperate regions tend to grow more vigorously as temperatures rise and CO₂ levels increase, highlighting the potential for climate change to exacerbate the frequency and severity of HABs in these regions. Gobler et al. (2017) found that ocean warming since 1982 has expanded suitable habitats for HABs in the North Atlantic and North Pacific oceans, primarily due to rising sea temperatures that have altered marine ecosystems and increased conditions favorable for these blooms. While controlled experiments and statistical analysis have shed light on some of these relationships, they are limited by their narrow range of variables and do not fully capture the complex interplay of real in situ factors (Yan et al., 2024b). The challenge of predicting the impact of climate change on HABs stems from the species-specific responses to interacting climate variables, such as temperature fluctuations and nutrient dynamics (Hallegraeff, 2010). This emphasizes the need for more comprehensive model simulations to integrate expansive environmental interactions. Predictive models, particularly those based on regression, are indispensable tools to bridge this knowledge gap.

Traditional approaches to predicting HABs, such as process-based and empirical statistical models, have provided foundational insights but often underperform due to the inability to account for complex environmental interactions or oversimplifications (Ralston and Moore, 2020; Xia et al., 2020; Yan et al., 2024b). Process-based models struggle with accurately representing biological processes due to the complexities of unknown algae dynamics (Deng et al., 2021; Flynn and McGillicuddy, 2018; Yu et al., 2021). While empirical-statistical models attempt to correlate environmental variables with HABs using formulas or statistical fits, the models frequently fail to capture complex nonlinear relationships and suffer from limited predictive accuracy (Franks, 2018; Wherry and Schenk, 2024; Yñiguez and Ottong, 2020). In contrast, machine learning models have emerged as a powerful alternative, offering advanced analytical capabilities to uncover complex, non-obvious patterns without requiring explicit mathematical formulations of the underlying processes (Bergen et al., 2019). Recent literature on predicting HABs using machine learning has predominantly incorporated target time lag features to enhance predictive precision (Deng et al., 2021; Wen et al., 2022). However, this reliance on including the target variable's autoregressive features to improve prediction presents a problem. While these models excel in forecasting, their dependence on known target values limits their utility for long-term climate change projections, as they cannot predict the future without access to future target variables. This limitation highlights the significant gap in research applying machine learning to explore the specific impacts of climate change on HABs, which requires urgent attention to understand and mitigate future ecological environmental challenges (Glibert, 2020). Bridging this gap requires innovative modeling approaches, which can accurately predict HABs without including target time lags in features.

To address the pressing need for effective water quality management in the bay, we identified several crucial gaps: 1) Current models often struggle to capture the complex interactions between multiple environmental factors that influence HABs. Additionally, these models frequently omit crucial variables due to data limitations, which diminishes their predictive power. To overcome these challenges, we will develop advanced machine learning models that integrate a broader range of environmental variables—such as temperature, precipitation, nutrient loads, water temperature, and specific humidity. By capturing these complex interactions, our models will provide a more comprehensive and accurate understanding of the factors driving HABs. 2) Autoregressive features (e.g., lagged chlorophyll-a values) can improve short-term forecasting—as shown in Liu et al. (2022) and Yan and Alamdari (2024) using hybrid models with temporal decomposition—but they rely on historical target values. This reliance introduces temporal dependencies that are not applicable for long-term projections under changing climate conditions, where future target values are unknown. For century-scale forecasting, these dependencies become problematic. Therefore, we developed non-autoregressive machine learning models that exclude past chlorophyll-a observations, ensuring that our projections are robust, climate-informed, and driven solely by exogenous environmental variables. 3) There is a need for robust climate projections that are bias-corrected and downscaled to local scales to improve predictive accuracy. Our research will utilize bias-corrected climate projections from multiple Global Climate Models (GCMs). This will ensure our predictions are grounded in reliable data that accurately reflect future climatic conditions. Our research aims to fill critical gaps in water quality management by developing advanced, interpretable machine learning models that integrate a wide range of environmental variables and utilize bias-corrected climate data. These models will enhance the accuracy of long-term HAB predictions, offering valuable insights for managing and mitigating HABs in North Biscayne Bay and similar regions.

This study builds on our previous work on machine learning algorithms for chlorophyll-a concentration predictions across Biscayne Bay (Yan et al., 2024a) using a larger dataset that includes multiple climate variables across 1–12-month lags for all features. The novelty of this study is the integration of bias-corrected climate projections with machine learning models that exclude autoregressive target variables, to predict HABs under future climate scenarios. Unlike previous studies that use short-term forecasting with autoregressive features, our approach allows for 2100 projections using climate-informed predictors only. To guide this effort, we have the

following research questions: (1) Can HABs in an urban coastal environment be predicted using machine learning models trained only on environmental variables without autoregressive inputs? (2) Which climate and hydrological drivers are most important for long-term HAB trends under climate change? (3) How will future climate scenarios and seasonal conditions (wet and dry seasons) influence the projected frequency and intensity of HABs in North Biscayne Bay? These questions advance the field by showing a new modeling path for long-term HAB forecasting in urban estuaries.

2. Methods and materials

2.1. Study area

North Biscayne Bay, downstream of the most intensely urbanized parts of Miami-Dade County, is an ecologically significant estuary sustaining rich biodiversity (Fig. 1). It serves as a crucial habitat for seagrass beds essential for maintaining water quality, providing food, and offering shelter to numerous marine species. However, this urbanized estuary has come under severe ecological stress due to anthropogenic influences and climatic shifts. Urban runoff from Miami-Dade delivers approximately 3130 kg/day of total nitrogen and 49 kg/day of total phosphorus to North Biscayne Bay, with canals contributing roughly 95 % of the total nitrogen and 90 % of the total phosphorus loads (Chin, 2020). These elevated nutrient inputs, along with high septic tank densities (over 45,000 units across the watershed), have been linked to persistent eutrophication symptoms and seagrass loss. Developed land cover in the watershed has also steadily increased, with chlorophyll-a concentrations rising at an average of 0.029 $\mu\text{g/L}$ per year from 1995 to 2014 (Millette et al., 2019). These combined anthropogenic pressures, compounded by climate change, have led to HAB occurrences and widespread seagrass degradation in recent years (Alexandre et al., 2021; Santos et al., 2020).

North Biscayne Bay is in a subtropical climate zone (Köppen climate classification: Cfa), with distinct wet and dry seasons. The area has an average annual temperature of 24.6°C, with summer highs often above 32°C and mild winters. Annual precipitation is around 1570 mm, with over 70 % of that in the wet season from June to October. Hydrologically, the bay receives inflow from a network of urban canals—primarily the C-8 Canal via the S29 discharge station—carrying stormwater and nutrient runoff from heavily developed

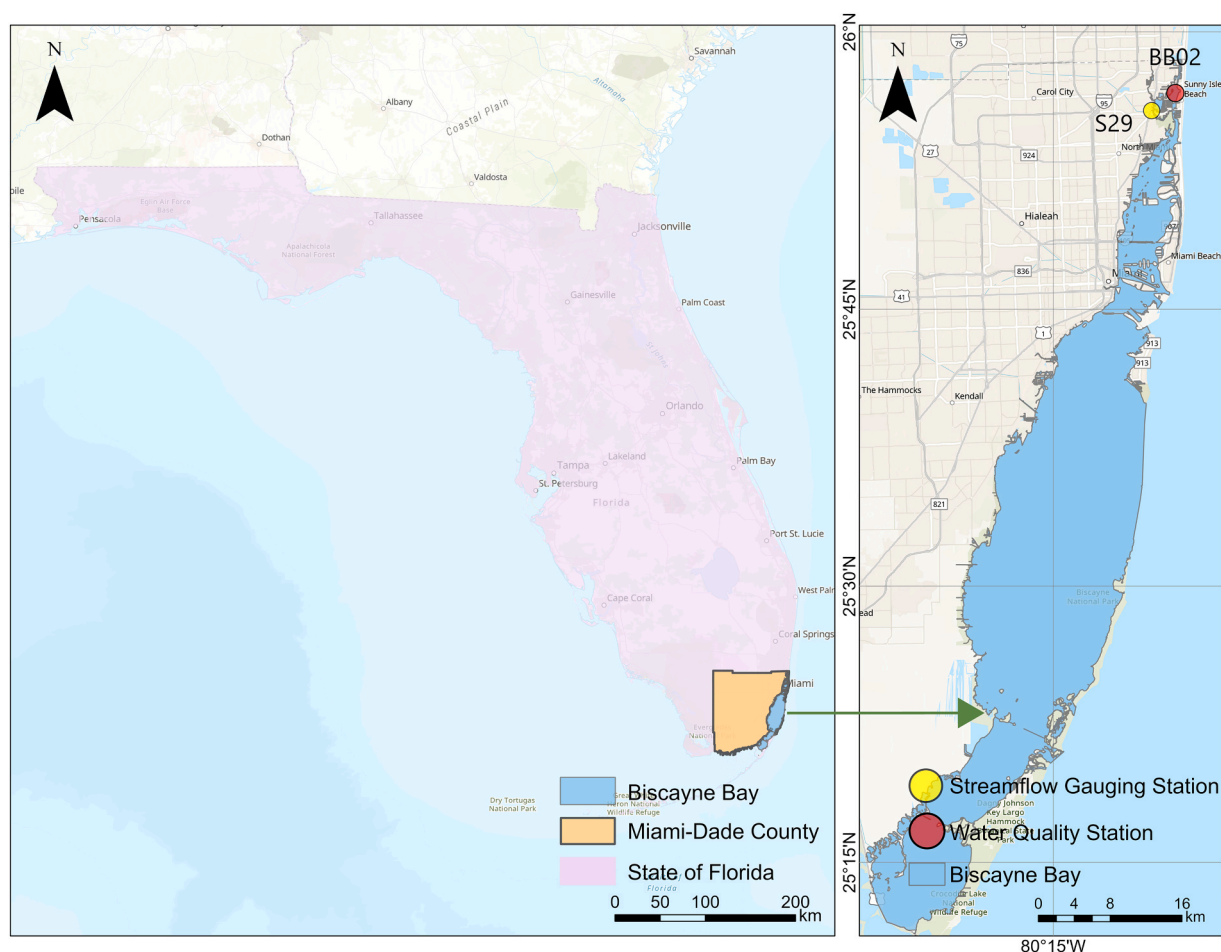


Fig. 1. The geographic location of water quality station BB02 and the nearest discharge station S29 to North Biscayne Bay.

upstream areas. This climatic and hydrological regime makes North Biscayne Bay prone to HABs, especially during the high-flow, high-temperature conditions of the wet season. The site with the most severe water quality issues, BB02, has been continuously monitored, yielding invaluable long-term datasets. This data is crucial for understanding the complex ecological processes influencing key ecological indicators such as chlorophyll-a concentrations, dissolved oxygen levels, and nutrient enrichment patterns, which form the basis of our predictive modeling efforts. The sensitivity of North Biscayne Bay to both local environmental management practices and global climate phenomena makes it an exemplary study area to explore the impacts of climate change on coastal ecosystems. Our study employs the wealth of historical in-situ water quality data from BB02 to develop our machine learning model, eventually aiming to project future states of HABs.

2.2. Data collection

The foundation of our study relies on rigorously quality-assured and quality-controlled (QA/QC) data. Chlorophyll-a, the sole target variable, is widely used as a proxy for phytoplankton biomass and an indirect indicator of potential HAB risk (Deng et al., 2021). Monthly datasets spanning various environmental parameters have been Thoroughly gathered, as detailed in Table 1, including terms, units, data sources, categories, and respective periods. Selection criteria for feature variables were informed by data availability and their demonstrated relevance to HAB influences in the literature review (Yan et al., 2024b). In enhancing the predictive robustness of our models, a 12-month lag was systematically incorporated for each feature, allowing the integration of temporal patterns to bolster predictive accuracy (Liu et al., 2022; Yan et al., 2024a).

For developing our machine learning models, all water quality data from station BB02 (1998–2020) was sourced from the Miami-Dade Division of Environmental Resources Management (DERM). This encompassed chlorophyll-a, ammonia nitrogen, NOx, dissolved oxygen, pH, turbidity, total phosphorus, and water temperature. Discharge data at station S29 was obtained from the South Florida Water Management District's (SFWMD) DBHYDRO database, proximal to BB02. Climate variables, including minimum, maximum, and air temperatures, specific humidity, wind speed, precipitation, and shortwave radiation flux, were provided by gridMET (Abatzoglou, 2013). The developed percent was computed from the National Land Cover Database (NLCD), leveraging SFWMD-provided watershed delineation used in our prior work and aggregating varying intensities of developed land divided by overall area to calculate a development metric (Yan et al., 2024a). Annual values for 1998–2020 were derived using linear interpolation and polynomial extrapolation from available data (2001, 2004, 2006, 2008, 2011, 2013, 2016, and 2019), assigning identical monthly values within each year. This approach was adopted due to the limited availability of continuous annual land use datasets, and our primary goal was to capture broad temporal trends rather than short-term fluctuations. Previous analysis in our study region

Table 1

Variables used in machine learning (ML) and quantile mapping (QM) development and application/projection.

Variable (unit)	Data source	Category	Time period
Chlorophyll-a (µg/L)	DERM	Target of ML development	1998–2020
		Target of ML application	2021–2100
Ammonia nitrogen (mg/L)	DERM	Feature of ML development	1998–2020
NOx (mg/L)		Feature of ML application*	2021–2100
Dissolved oxygen (mg/L)			
pH			
Turbidity (NTU)			
Total phosphorus (mg/L)			
Water temperature (°C)	DERM	Feature of ML development	1998–2020
		Feature of ML application**	2021–2100
Discharge (cfs)	DBHYDRO	Feature of ML development	1998–2020
		Feature of ML application**	2021–2100
Minimum temperature (°C)	gridMET	Feature of ML development	1998–2020
Maximum temperature (°C)		Feature of ML application***	2021–2100
Air temperature (°C)		Observed historical data for QM development	1979–2014
Specific humidity (kg/kg)			
Wind speed (m/s)			
Precipitation (kg/m ²)			
Shortwave radiation flux (W/m ²)			
Developed percent (%)	NLCD	Feature of ML development	1998–2020
		Feature of ML application****	2021–2100
Minimum temperature (°C)	NEX-GDDP-CMIP6	Simulated historical data for QM development	1979–2014
Maximum temperature (°C)		Simulated projected data for QM bias-correction	2015–2100
Air temperature (°C)			
Specific humidity (kg/kg)			
Wind speed (m/s)			
Precipitation (kg/m ²)			
Shortwave radiation flux (W/m ²)			

Note: * indicates the data in that future 2021–2100 period is constructed by repeating the observed 2001–2020 sequence five times. ** indicates the data in that future 2021–2100 period is predicted from climate variables using multiple linear regression. *** indicates the data in that future 2021–2100 period is bias-corrected from the NEX-GDDP-CMIP6 dataset using quantile mapping. **** indicates the data in that future 2021–2100 period is extrapolated by linear regression.

(Yan et al., 2024a) has shown that land use change in the watershed follows a relatively smooth long-term trajectory, supporting the use of linear interpolation and low-order polynomial extrapolation. While this method introduces some uncertainty in representing interannual variability, it provides a reasonable approximation for long-term modeling. Future work could incorporate higher-resolution remote sensing or dynamically modeled land use data to fill current data gaps and refine these projections.

Since there are no publicly available high-resolution future water quality projections for Biscayne Bay, we assumed the seasonal distribution of water quality variables (e.g. nutrient concentrations) observed from 2001 to 2020 would persist through 2100. This stationarity assumption does not capture changes due to land use or regulatory changes but allows for a conservative baseline to isolate the effects of climate-driven meteorological and hydrological factors on chlorophyll-a dynamics. Replicating observed records avoids making predictions from climate inputs which could introduce more uncertainty and reduce long-term model reliability (Zhi et al., 2023). Climate projections were downscaled from the NEX-GDDP-CMIP6 dataset using quantile mapping, providing refined estimates that reflect future climatic conditions (Thrasher et al., 2022). Specifically, we selected the BCC-CSM2-MR, CNRM-ESM2-1, MPI-ESM1-2-LR, and MRI-ESM2-0 GCM models for their demonstrated precision in capturing climatic patterns pertinent to Southeast Florida (Rahimi et al., 2024). Future discharge and water temperature were estimated using multivariate linear regression models trained and validated on historical climate variables, with inputs derived from bias-corrected climate projections. Despite their simplicity, these models achieved high predictive performance (Table 2, $R^2 > 0.8$, low MAE and MAPE), supporting their adequacy for this application. While our assumption of stationary water quality inputs provides a conservative baseline to isolate climate-driven changes, it cannot capture potential nutrient shifts under changing land use or other anthropogenic influences. Future work should integrate dynamic nutrient models (e.g., SWAT) once calibration data become available, to better represent these processes. Similarly, our use of linear regression to estimate discharge and water temperature captures first-order trends but omits potential threshold behaviors; future efforts could benchmark against more mechanistic or nonlinear approaches. The developed percent metric was extrapolated linearly from 2006 to 2019 data to estimate future development scenarios, as the data post-2006 showed no abrupt changes and remained relatively stable. With the assembly of the 2021–2100 feature dataset, our machine learning model received these inputs to predict chlorophyll-a concentrations, thus illuminating the potential future trends of HABs in North Biscayne Bay.

2.3. HAB predictive model and climate model bias-correction development and evaluation

Our study's machine learning dataset was divided into a training set for 1998–2015 and a testing set for 2016–2020. We employed three machine learning models: Support Vector Machine (SVM), Random Forest (RF), and Multi-layer Perceptron (MLP). We chose SVM, RF, and MLP as they are different and widely used classes of machine learning algorithms for nonlinear regression with multivariate environmental datasets. SVM is good for small-to-medium datasets with nonlinear relationships (Hearst et al., 1998); RF is robust to overfitting and provides ensemble-based decision making (Breiman, 2001); MLP is a type of artificial neural network that can capture complex high-dimensional interactions (Rumelhart et al., 1985). By evaluating these diverse algorithms, we covered kernel-based, ensemble-based and neural-network-based approaches. These machine learning models were compared using the coefficient of determination (R^2), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) to ensure the robustness and predictive accuracy of the models for both training and unseen testing data. All these models were subject to hyperparameter tuning via grid search and feature selection through forward selection and backward elimination to enhance performance. To statistically compare the predictive performance of the models, we applied the Diebold–Mariano (DM) test (Diebold and Mariano, 1995) on the out-of-sample predictions (2016–2020). The DM test evaluates whether the prediction errors from two models differ significantly, providing a rigorous statistical basis for model selection. Subsequently, the optimal models were used to predict chlorophyll-a concentrations for 2021–2100 using the future climate dataset. To improve model interpretability, we used SHAP (SHapley Additive exPlanations), a unified framework based on cooperative game theory that quantifies each feature's contribution to a model's prediction (Lundberg et al., 2020). SHAP values were used to rank the most important environmental variables affecting predicted chlorophyll-a concentrations. SHAP was performed with 'shap 0.43.0' (Lundberg et al., 2020; Lundberg and Lee, 2017).

Bias correction is essential for aligning coarse-scale GCM outputs with local observational records, particularly in regions like Southeast Florida where fine-scale microclimatic variations can substantially influence water quality outcomes. By applying quantile mapping, we ensure that the statistical properties of the climate inputs (mean, variance, and distribution) closely match those of the observed records, which is critical for generating reliable machine learning predictions of chlorophyll-a concentrations. The bias-corrected future climate dataset was accomplished using Quantile Mapping (QM), which established a statistical relationship between the observed climate dataset (gridMET) and the CMIP6 simulated historical climate dataset for 1979–2014. This method

Table 2

Metrics of machine learning HAB (chlorophyll-a, Chl-a) predictive results and metrics of multiple linear regression water temperature (WT) and discharge (DC) predictive results.

Metrics	SVM Chl-a	RF Chl-a	MLP Chl-a	MLR WT	MLR DC
R^2 for train	0.82	0.80	0.64	0.89	0.83
R^2 for test	0.74	0.62	0.66	0.91	0.83
MAE for train	0.69	0.70	0.99	0.92	83.20
MAE for test	0.92	1.18	1.12	0.71	105.92
MAPE for train	0.33	0.34	0.48	0.04	0.16
MAPE for test	0.26	0.33	0.32	0.03	0.31

allowed us to bias-correct the CMIP6 future climate projections, including different shared Socioeconomic Pathways (SSPs), SSP126 (mild), SSP245 (moderate), and SSP585 (extreme), to a local scale, resulting in finer climate datasets for future machine learning prediction applications (Pastén-Zapata et al., 2020). To evaluate the effectiveness of the quantile mapping bias-correction method applied to GCM outputs we used three non-parametric tests that together assess variance, distribution and median equivalence between the downscaled simulated data and observed climate data. Specifically: (1) the Kolmogorov-Smirnov (KS) Test was used to compare the empirical cumulative distribution functions (ECDFs) of downscaled and observed datasets, detecting overall distributional differences based on the maximum difference between their cumulative distributions (Smirnov, 1948); (2) the Mann-Whitney U (MWU) Test was applied to compare the medians of two independent samples by ranking and summing their values, making it suitable for non-normally distributed or ordinal climate variables (Mann and Whitney, 1947); and (3) Levene's Test was used to assess variance homogeneity by analyzing the absolute deviations from group means, ensuring that the downscaled data preserved the variability characteristics of observed records (Levene, 1960). These tests are well-suited for environmental time series data, which often exhibit non-normal distributions, temporal dependencies, and heterogeneous variances. The KS Test is non-parametric and robust against deviations from normality, enabling a full comparison of the observed and simulated distributions. The MWU Test detects significant differences in median values without assuming normality, making it suitable for datasets with outliers. The Levene's Test evaluates variance homogeneity, which is important for assessing the spread and uncertainty of climate model outputs. Together, these tests confirm that the bias-correction process preserves key statistical properties—distribution, central tendency, and variability—of historical climate data, thereby ensuring the robustness of bias-corrected climate inputs for long-term chlorophyll-a prediction models. This directly supports our research objective of producing statistically comparable, median-preserving, and variance-consistent future climate datasets for HAB projections in North Biscayne Bay. A high P -value suggests no significant difference between the simulated and observed historical datasets, which affirms the reliability of our downscaling method.

All modeling procedures were implemented in a Python 3.8.8 environment. Machine learning regression models were called from the 'sklearn 1.1.3' library, while the quantile mapping was executed using 'python-cmethods 2.2.1' (Pedregosa et al., 2011; Schwertfeger et al., 2023). The ECDF assessment was conducted with 'statsmodels 0.13.5' and 'scipy 1.10.1' (Seabold and Perktold, 2010; Virtanen et al., 2020). The development details of all models can be found in Fig. 2.

2.3.1. Support vector regressor

The SVM model is optimized for regression tasks and implemented with a radial basis function (RBF) kernel. The hyperparameters,

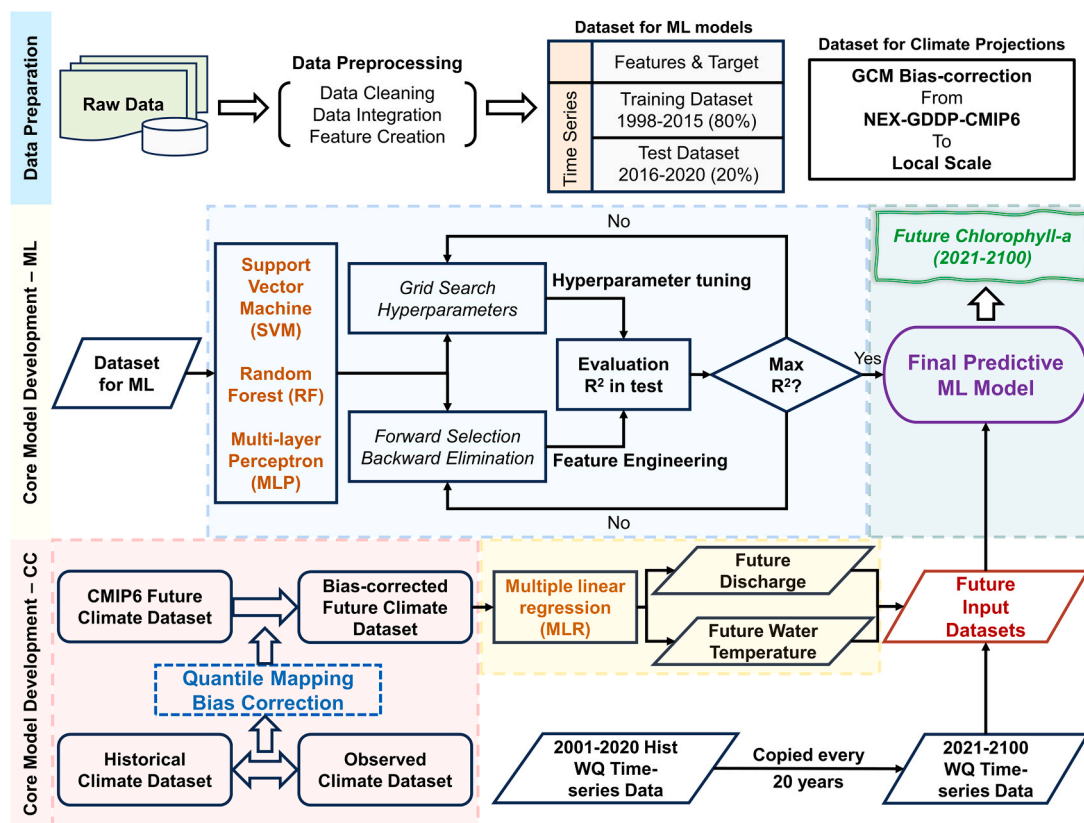


Fig. 2. The flowchart illustrating the methodology for predictive modeling of future chlorophyll-a concentration in North Biscayne Bay, including data preparation, core machine learning model development and application, and bias-correction downscaling model development.

such as cost C and gamma γ , are tuned to minimize generalization error. SVM regression attempts to fit the error within a certain threshold and is expressed by:

$$f(x) = \sum_{i=1}^n (\alpha_i - \alpha_i^*) \bullet K(x_i, x) + b \quad (1)$$

where α_i and α_i^* are the Lagrange multipliers, K is the kernel function, and b is the bias.

2.3.2. Random forest regressor

The RF regressor integrates multiple decision trees to address the overfitting problem common in individual trees. The model's hyperparameters, including the number of trees n_{trees} and maximum depth d_{max} , are fine-tuned for accuracy. The RF model is described as:

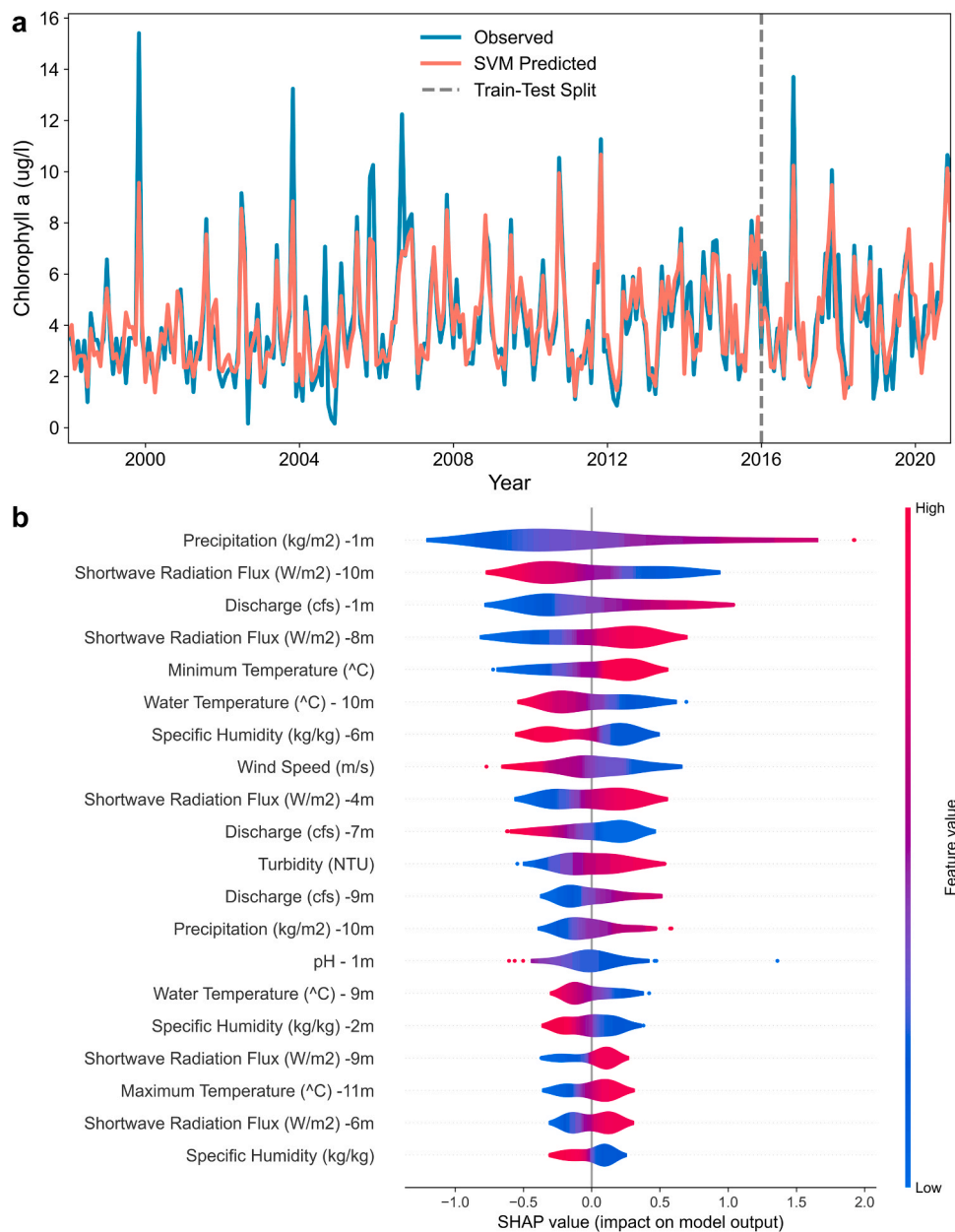


Fig. 3. Visualization of the best HAB predictive model (SVM) in time series and feature importance. a shows the temporal comparison of observed against SVM-predicted chlorophyll-a concentrations during the training period (1998–2015) and testing period (2016–2020).

$$f(x) = \frac{1}{n_{trees}} \sum_{i=1}^{n_{trees}} T_i(x) \quad (2)$$

where T_i represents the i -th decision tree.

2.3.3. Multi-layer perceptron regressor

MLP is a feedforward artificial neural network (ANN) structured with multiple layers, utilizing backpropagation for training. The model's architecture is calibrated to optimize prediction, including the number of hidden layers H and activation function ϕ . The MLP function is generalized as follows:

$$f(x) = \phi(W \bullet x + b) \quad (3)$$

where W and b are weights and biases, and ϕ is the activation function.

2.3.4. Quantile mapping

QM is used for bias correction in climate models, aligning the distribution of the simulated data with that of observed data. This is conducted by mapping the model output's quantiles to the observed data's quantile to correct systematic model biases. The general expression for QM is:

$$F_{model}^{-1}(F_{obs}(x)) \quad (4)$$

where F_{model}^{-1} is the inverse cumulative distribution function (CDF) of the simulated data, and F_{obs} is the CDF of the observed data.

3. Results

3.1. Development of HAB predictive models using machine learning

Developing predictive models for HABs in North Biscayne Bay used a data-driven approach, employing machine learning algorithms to process historical observed water quality and climate data. Our modeling efforts were grounded on the established premise that chlorophyll-a concentration serves as a widely used proxy for phytoplankton biomass and an indicator of potential HAB risk (Deng et al., 2021). The models were trained on the dataset from 1998 to 2015 and tested on data from 2016 to 2020. The hyperparameter tuning results are provided in Table S1. Evaluation metrics showed the SVM model's superior performance with a training R^2 of 0.82, indicating a high level of variance explained by the model, and a test R^2 of 0.74, affirming its predictive validity on unseen data. This model also maintained lower MAE across training (0.69) and testing (0.92) phases, compared to the RF and MLP models. MAPE results followed a similar trend, with SVM outperforming its counterparts on training (0.33) and testing (0.26) data, reflecting a consistent and precise predictive capability across temporal scales (Table 2). The temporal comparison of observed versus SVM-predicted chlorophyll-a concentrations demonstrated the model's predictive ability to closely mimic the data trends and patterns across the training and testing periods, despite the SVM model not capturing some of the extreme values observed in the data (Fig. 3a). The DM tests indicated that the SVM model had lower prediction errors than RF and MLP (negative DM values indicate smaller errors for SVM), with differences approaching statistical significance ($p < 0.10$; SVM vs RF: DM = -1.80, $p = 0.077$; SVM vs MLP: DM = -1.92, $p = 0.060$), supporting its selection as the primary predictive model. Overall, the results indicated that the model generally provides accurate predictions.

An intricate SVM model's interpretability was elucidated through the SHAP values, which detailed the influence of each environmental feature on model predictions. Based on the SHAP analysis in Fig. 3b, the top 20 features impacting the SVM model's predictions revealed significant insights into the conditions influencing chlorophyll-a concentrations. Focusing on the current and 1-month lag features, which offered the most direct insight into HAB prediction, we identified key environmental variables that notably impacted algal blooms. Precipitation, at a 1-month lag, emerged as the most influential predictor, showing a strong positive association with chlorophyll-a levels. Similarly, upstream discharge with a 1-month lag and the current month's minimum temperature positively influenced chlorophyll-a. Conversely, the current month's wind speed negatively impacted chlorophyll-a levels.

In conclusion, the predictive models showcased the effectiveness of machine learning in identifying and predicting chlorophyll-a concentrations. The inclusion of SHAP analysis allowed for model interpretability and provided valuable insights into the significant features driving the predictions.

b shows SHAP (SHapley Additive exPlanations) values interpreting each feature's relative positive or negative importance in the SVM model.

3.2. Bias-correction and validation of Global Climate Models (GCMs)

Our study used bias-corrected GCMs to generate reliable climate projections by Quantile Mapping (QM), which helped align the distributions of the simulated historical climate data with observed historical records. The NEX-GDDP-CMIP6 dataset provided a basis for downscaling climate projections, covering mild to extreme climate scenarios (SSP126, SSP245, and SSP585) and generating future climate datasets suitable for regional application.

Various statistical tests were used to validate the bias-correction process for homogeneity between observed and bias-corrected simulated historical climate data, including Levene's test for variance, the Kolmogorov-Smirnov test for distribution, and the Mann-Whitney *U* test for median. The high *P*-values (close to 1) obtained from these statistical tests shown in Table 3 indicated insufficient evidence to reject the null hypothesis and no significant difference in variance, distribution, and median between the simulated and observed datasets, confirming the effectiveness of the bias correction. The bias-corrected GCMs produced trustworthy future climate datasets, supporting robust chlorophyll-a machine learning predictions in North Biscayne Bay. The reliability of these projections provides confidence in their application to predict future trends of HABs in the context of changing climatic conditions.

Fig. 4 depicts observed historical and bias-corrected future climate variables, illustrating their trends and distribution. Minimum temperature, air temperature, and specific humidity exhibit upward trends in historical and projected scenarios, with SSP585 showing the most severe slope, and SSP126 being the mildest. Surprisingly, maximum temperature has shown a slight decline trend in historical data, but projected scenarios show an increasing trend. For wind speed, historical data shows a decreasing trend, and all three future climate scenarios exhibit similar declining trends. The historical trend is increasing for precipitation, but the future projections contradict this trend, showing a downward slope, particularly steep in SSP585. Lastly, while shortwave radiation has historically increased slightly, future projections suggest a slight decrease.

3.3. Projections of HABs in North Biscayne Bay under future climate scenarios

In assessing the potential future impacts of climate change on HABs in North Biscayne Bay, our study leveraged advanced predictive modeling developed in Section 3.1 and bias-corrected climate data in Section 3.2 to project the future chlorophyll-a concentrations up to 2100.

Fig. 5 illustrates the variability in chlorophyll-a concentration prediction from different GCMs across the SSP scenarios. The standard deviation (SD) measures variability among models, indicating uncertainty in future chlorophyll-a projections. SSP 126 has the highest variability, with a mean SD of 0.624, ranging from 0.054 to 1.753, and a median of 0.571. This suggests that under SSP 126, a relatively conservative future pathway, different GCMs show more disagreement, indicating higher uncertainty in predicting future chlorophyll-a concentrations. SSP 245 has lower variability than SSP 126, with a mean SD of 0.602 and a median of 0.561. SSP 585 has the least variability, with a mean SD of 0.588 and a median of 0.537, suggesting the highest model agreement despite representing the most extreme pathway. Comparing this with the observed data range (0.16–15.41), the observed data's variability is much higher than the predictive SD from different GCMs across the SSP scenarios. This suggests the four GCM models show similar patterns in projecting the future for all SSPs.

Fig. 6a provides a compelling visualization of the observed and predicted time series and trends in chlorophyll-a concentrations up to 2100, integrating the predictions generated from the SVM model using averaged outputs from four GCM models, BCC-CSM2-MR, CNRM-ESM2-1, MPI-ESM1-2-LR, and MRI-ESM2-0, under SSP126, SSP245, and SSP585 scenarios. Each trend line is accompanied by a 95 % confidence interval, which widens substantially toward the end of the century, with the largest uncertainties observed in 2100 across all scenarios (CI widths: 0.323 $\mu\text{g/L}$ for SSP126, 0.324 $\mu\text{g/L}$ for SSP245, and 0.311 $\mu\text{g/L}$ for SSP585), reflecting compounding uncertainties in model projections and climate variability. The observed data up to 2020 show fluctuations with numerous extreme

Table 3

P-values for comparing the statistical properties of Empirical Cumulative Distribution Functions (ECDF) between downscaled historical Global Climate Model (GCM) data and observed climate records. Statistical tests include Levene's Test for homogeneity of variances, the Kolmogorov-Smirnov (KS) Test for ECDF distribution differences, and the Mann-Whitney *U* (MWU) Test for discrepancies in the median. The GCM datasets evaluated are from BCC, CNRM, MPI, and MRI models.

Feature	Statistical test	BCC <i>P</i> -value	CNRM <i>P</i> -value	MPI <i>P</i> -value	MRI <i>P</i> -value
Minimum Temperature	Levene's Test	0.9925	0.9988	0.9903	0.9957
	KS Test	1.0000	1.0000	1.0000	1.0000
	MWU Test	0.9751	0.9718	0.9788	0.9964
Maximum Temperature	Levene's Test	0.9942	0.9999	0.9911	0.9917
	KS Test	0.9963	0.9774	0.9898	0.9898
	MWU Test	0.9990	0.9849	0.9788	0.9755
Air Temperature	Levene's Test	1.0000	0.9992	0.9988	0.9977
	KS Test	1.0000	1.0000	1.0000	1.0000
	MWU Test	0.9999	0.9982	0.9955	1.0000
Specific Humidity	Levene's Test	0.9844	0.9983	0.9949	0.9922
	KS Test	1.0000	1.0000	1.0000	1.0000
	MWU Test	0.9947	0.9997	0.9968	0.9925
Wind Speed	Levene's Test	0.9999	0.9902	0.9982	0.9962
	KS Test	1.0000	1.0000	1.0000	1.0000
	MWU Test	0.9990	0.9953	0.9910	0.9992
Precipitation	Levene's Test	0.9995	0.9973	0.9985	0.9987
	KS Test	1.0000	1.0000	1.0000	1.0000
	MWU Test	0.9986	0.9992	0.9975	0.9960
Shortwave Radiation Flux	Levene's Test	0.9999	0.9915	0.9963	0.9987
	KS Test	1.0000	1.0000	1.0000	1.0000
	MWU Test	0.9958	0.9995	0.9975	0.9942

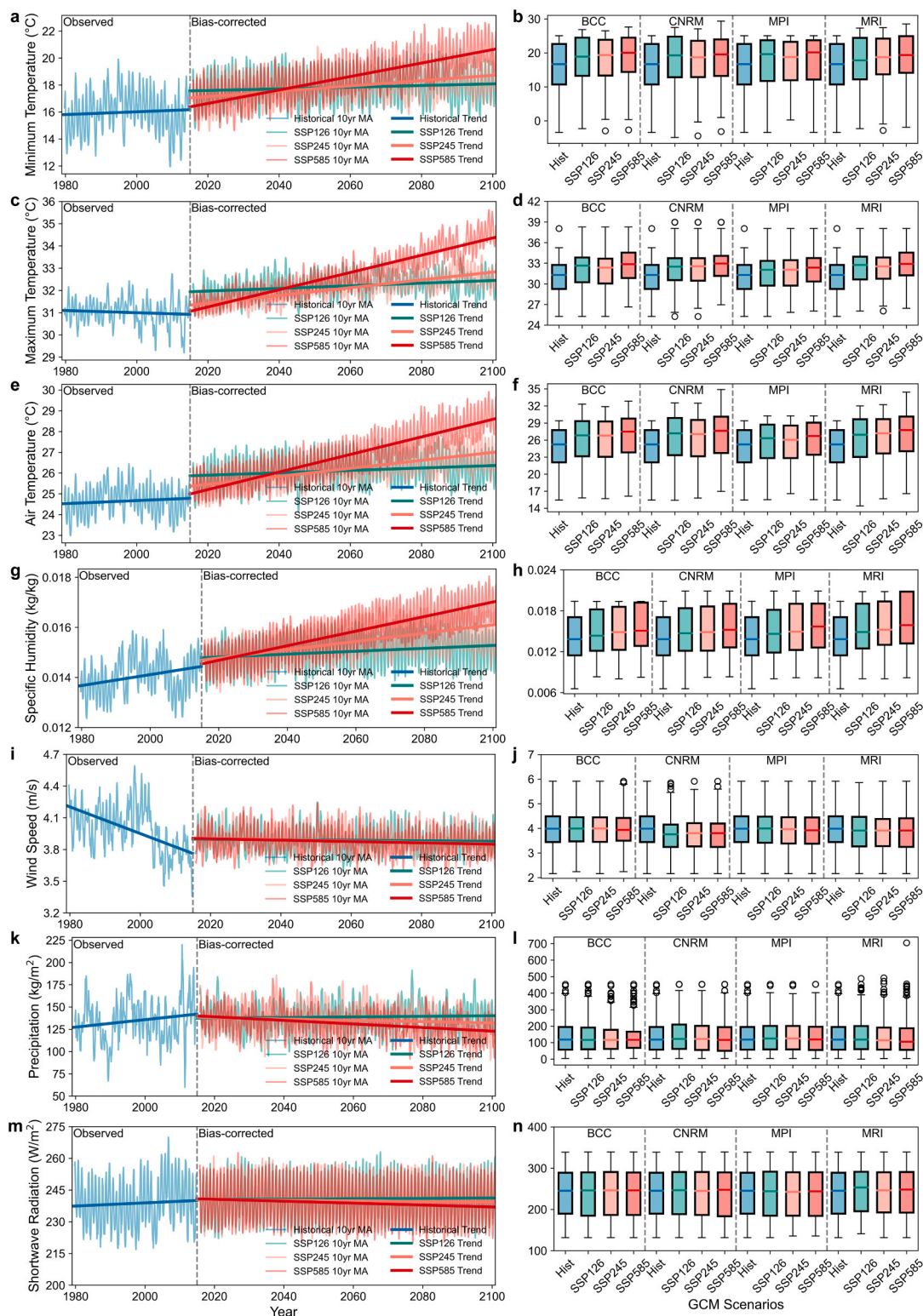


Fig. 4. Observed historical climate variables and future climate variables after bias-correction. The seven left plots depict the decadal averages and trends for all climate variables, comparing observed historical data with bias-corrected future averages from four climate models (BCC-CSM2-MR, CNRM-ESM2-1, MPI-ESM1-2-LR, and MRI-ESM2-0) under each climate scenarios SSP126, SSP245, and SSP585. The seven right plots present box plots comparing observed historical data with bias-corrected future outputs from the same four models and three scenarios, illustrating the distribution of each variable.

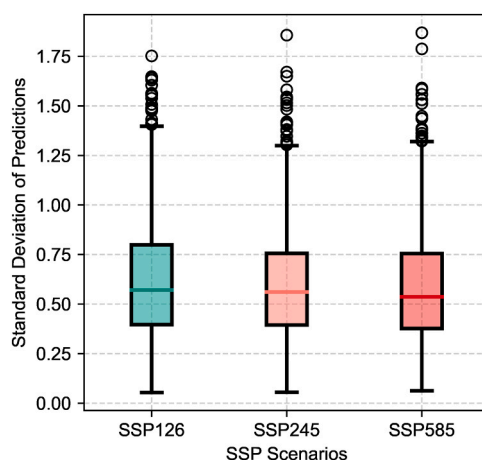


Fig. 5. The standard deviation of chlorophyll-a predictions from different global climate models (GCMs) under various Shared Socioeconomic Pathway (SSP) scenarios.

values, characterized by a clear upward trend with a slope (k) of 0.0045. The subsequent future projections, while also demonstrating an overall upward trend in chlorophyll-a concentrations, do not exhibit extremes as pronounced as those in the observed historical data. Under the mildest climate scenario, SSP126, the increase in chlorophyll-a concentrations is substantial, suggesting that even under less severe climate conditions, the increasing concentrations trend prevails. However, in the more severe scenarios, SSP245 ($k = 0.0007$) and SSP585 ($k = 0.0005$) exhibit slower trends in chlorophyll-a concentrations. Trends associated with each scenario and their respective P -values affirm the statistical significance of the predicted changes.

Fig. 6b illustrates the seasonal patterns in projected chlorophyll-a concentrations, distinguishing between wet and dry seasons based on the climate of South Florida, where the wet season spans from June to October and the dry season from November to May (Rahimi et al., 2024). After obtaining the future projected chlorophyll-a levels in Fig. 6a, we averaged the values for each season and categorized them into short-term and long-term projections. The results show that wet season concentrations almost consistently exceed those in the dry season, with a marked increase in chlorophyll-a levels over the long term compared to the short term. Specifically, short-term projections indicate an average chlorophyll-a concentration of 4.04 $\mu\text{g/L}$ during the wet season and 3.29 $\mu\text{g/L}$ during the dry season, while long-term projections show an increase, with 4.61 $\mu\text{g/L}$ for the wet season and 3.52 $\mu\text{g/L}$ for the dry season.

4. Discussion

4.1. Efficacy of machine learning algorithms in HAB prediction

Considering the diversity, interactions, and complexity of the various physicochemical and biological factors that drive algal bloom dynamics, developing a robust predictive model is challenging (Xia et al., 2020). Compared to traditional empirical and process-based models, machine learning methods provide greater flexibility in capturing nonlinear environmental interactions (Bergen et al., 2019; Zhong et al., 2021). Studies such as Zhou et al. (2024) and Chen et al. (2018) have demonstrated their utility in environmental applications based on complex datasets. Our model builds on and advances previous research in estuarine water quality prediction by combining machine learning with bias-corrected climate projections, offering both improved interpretability and long-term forecasting capacity. Carey et al. (2011) showed that watershed land use metrics and nutrient inputs predict nutrient variability, but lacked climate predictors and insight into nonlinear contributions. Our framework addresses this gap by including climate-informed predictors (precipitation, temperature, wind speed) and SHAP-based interpretability, achieving predictive accuracy and mechanistic understanding of chlorophyll-a variability. Applying machine learning in HAB prediction enhances monitoring and management capacity (Ly et al., 2021). These models often leverage the autoregressive features of target variables from previous time steps to enhance predictive accuracy (Deng et al., 2021).

Unlike many HAB prediction studies, we excluded autoregressive features (e.g., lagged chlorophyll-a values). While these features can improve short-term forecasts, they are unsuitable for century-scale projections where future target values are unknown. This design choice ensured that our model remained robust and climate-informed rather than reliant on unavailable autoregressive data. Among the three models tested, the SVM exhibited superior performance (Table 2), consistent with its strength in handling nonlinear relationships (Roy and Chakraborty, 2023). However, as highlighted in recent work (Bhayani et al., 2023; Zhu et al., 2023), the quality and availability of input data often play a more critical role in prediction performance than algorithm choice. This reinforces the need for sustained investment in high-quality climate and water quality monitoring to support reliable HAB prediction.

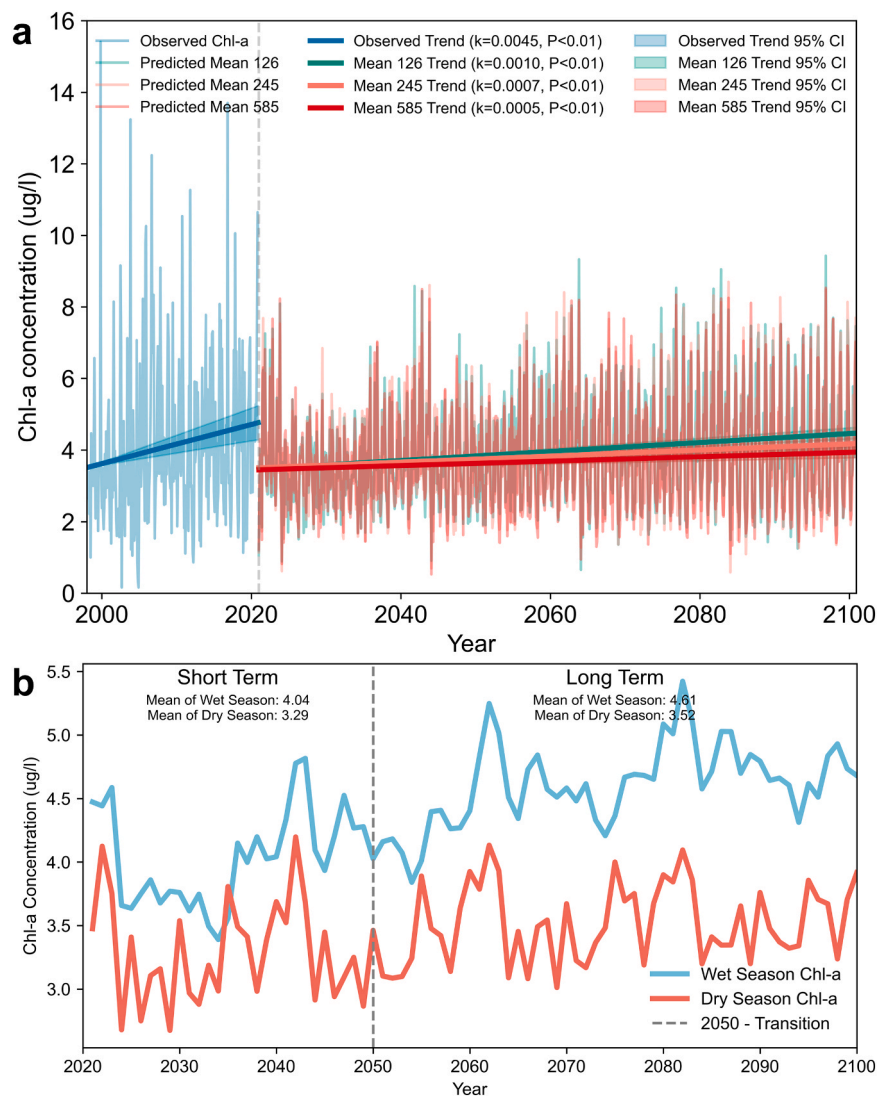


Fig. 6. Future projected time series of chlorophyll-a concentrations under different climate scenarios and wet and dry seasons. **a** Time series and trends of observed and predicted chlorophyll-a concentrations. The observed chlorophyll-a shows trends until 2020, with the subsequent predictions using the SVM model in SSP126, SSP245, and SSP585 by four averaged GCM models extending to 2100. Each trend line represents the trend of the prediction results in chlorophyll-a concentrations over time, along with 95 % confidence intervals (CI). Trend values (k) and P -values (P) are provided for each time series. The vertical dashed line marks the division between observed and predicted data. **b** Projected time-series chlorophyll-a concentrations averaged over all models and shown for both wet and dry seasons.

4.2. Identification of important features in HAB prediction

An integral component of our analysis was the evaluation of feature importance using SHAP values (Lundberg and Lee, 2017). We concentrated on features from the current and previous month, reflecting the typical growth cycle of harmful algae such as dinoflagellates and diatoms (Yan et al., 2024b). Features with longer lags were excluded due to their limited interpretability. Our analysis highlighted precipitation and discharge from one month prior, together with the current month's minimum temperature and wind speed, as the most influential predictors (Fig. 3b). These findings align with previous research indicating that precipitation and discharge deliver nutrients and organic matter that stimulate algal growth (Sinha et al., 2017; Sun et al., 2022). The strong influence of precipitation and discharge on HABs in our model highlights the pronounced hydrological sensitivity of North Biscayne Bay (Fig. 3). In South Florida, the wet season (June–October) is characterized by intense rainfall that drives stormwater runoff and managed canal discharges—especially from infrastructure like the S29 control structure—into the bay. These hydrological pulses mobilize and transport nutrients and organic matter from surrounding landscapes, creating conditions for phytoplankton growth and HABs. The one-month lag between precipitation/discharge and elevated chlorophyll-a in our model captures the time it takes for nutrients to be transported, accumulated and for algae to grow. This pattern is consistent with the “pulse-shunt” concept whereby rainfall pulses

deliver nutrients to estuarine zones and stimulate algal growth. This connection between seasonal hydrology and HAB formation is more pronounced in highly urbanized watersheds like North Biscayne Bay where canal discharges create strong nutrient gradients that directly impact nearshore water quality. Our results are consistent with [Caccia and Boyer \(2005\)](#) and [Carey et al. \(2011\)](#) who showed that urban and agricultural land uses increase nutrient delivery to estuarine systems via canal networks and contribute to eutrophication and bloom risk.

Interestingly, although nitrogen and phosphorus were included as predictors, their SHAP values were lower than precipitation and discharge. This indicates that hydrological variables act as integrative proxies—not only capturing episodic nutrient loading but also transporting other constituents such as silicon, iron, and trace metals that influence algal dynamics. The strong positive associations suggest that short-term increases in rainfall and discharge substantially promote bloom development. It shows that regional hydrology is the main driver of chlorophyll-a and supports using precipitation and discharge as predictors in HAB models. This means hydrologically-informed management during the wet season.

Lower wind speeds may reduce vertical mixing and dispersion of algal cells ([Deng et al., 2018](#); [Ralston et al., 2015](#)). This allows smoother algal vertical migration, optimizing light and nutrient conditions for bloom development ([Ranjbar et al., 2022](#); [Xue et al., 2022](#)). Additional important environmental factors included the current month's turbidity, which often rises due to HAB occurrence ([Wang et al., 2019](#)), the last month's pH preference for more acidic conditions in Biscayne Bay ([Yan et al., 2024a](#)), and the current month's specific humidity, which correlates with higher temperatures promoting algal growth. This convergence of evidence from our study and existing research reinforces the validity of these features as reliable indicators of HABs. Also, it suggests that our predictions are trustworthy and can be used for future climate change simulations. However, as [Ralston and Moore \(2020\)](#) have suggested, process-based models are often more trusted, whereas machine learning models may suffer from extrapolation issues. Therefore, this innovative ML-based predictive application for future climate projection should be not only based on the favorable metrics of the prediction outcomes ([Table 2](#)) but also on a rational analysis of feature importance that aligns with other relevant literature. Our SHAP results further suggest that small shifts in climate drivers—such as lower wind speed—and discharge pulses can interact to amplify bloom risk; this is operationally relevant in a managed-canal system like North Biscayne Bay where seasonal discharge can be adjusted.

However, it is essential to recognize the limitations in our feature selection in the predictive model. Due to data availability, our models did not include potentially influential variables like biological factors (algal predators), stratification, and salinity levels. Addressing these gaps in future work will improve model accuracy and generalizability ([Wells et al., 2020](#)).

4.3. Influences of climate variable adjustments post-bias-correction

Our results show the quantile mapping bias-correction method effectively adjusted GCM data distributions to match observed climate variables, as supported by the high *P*-values from Levene's, KS, and MWU tests ([Table 3](#)). This correction reduces mismatches and variance distortions, making long-term HAB forecasts more reliable under different climate scenarios.

The future climate scenarios in [Fig. 4](#) reveal key post-bias-correction trends. Minimum, maximum, and air temperatures, along with specific humidity, all show upward trends, with SSP585 steepest and SSP126 most moderate. This warming trend aligns with historical observations, except for maximum temperature, which declined slightly in the past but is projected to rise. Wind speed consistently declines historically and across scenarios. Solar radiation, historically slightly increasing, is projected to decrease modestly. The projected warming and reduced wind may promote algal growth in North Biscayne Bay ([Figs. 3b, 4](#)).

Wind speed decline has important implications for algal bloom dynamics in subtropical estuaries like North Biscayne Bay. Lower wind can suppress mixing, enhance stratification, and increase light availability—conditions favorable for phytoplankton ([Wells et al., 2015](#)). Our SHAP analysis showed a negative relationship between wind speed and chlorophyll-a concentrations, supporting the idea that lower wind speeds contribute to bloom development by promoting water column stability ([Fig. 3b](#)). In North Biscayne Bay, strong canal discharges already promote stratification, and reduced wind can prolong these bloom-favorable conditions ([Langevin, 2001](#)). Yet wind effects are context-dependent: moderate mixing may stimulate algal growth by resuspending nutrients ([Heil and Muni-Morgan, 2021](#)), while strong winds can disrupt stratification and disperse blooms ([Li et al., 2023](#)). These nonlinear effects highlight the importance of understanding local stratification when interpreting wind impacts. Solar radiation shows an ambiguous effect. Decreases could reduce photosynthesis, but some studies link HABs to periods of reduced radiation, suggesting complex and sometimes bloom-favoring interactions ([Vianna et al., 2023](#)).

A critical focus of our study is the precipitation due to its prominent role as the most influential feature in our feature importance analysis and its strong positive correlation with discharge, another key predictor of chlorophyll-a concentration ([Fig. 3b](#)). Historically, precipitation increased, which contradicts the future projections where a decreasing trend is noted across all SSP scenarios, steepest in SSP585 ([Fig. 4k](#)). This trend is particularly notable as it implies a potential reduction in nutrient runoff essential for algal growth ([Glibert, 2020](#)), thus impacting our predictive models for future chlorophyll-a. This also helps explain the counterintuitive finding that SSP126 shows a steeper increase in chlorophyll-a than SSP585. As SHAP shows, precipitation and discharge are strong positive predictors; under SSP585, their decline suppresses bloom drivers. Because SSP585 projects the largest decline in precipitation and consequently reduced discharge (which is derived from climate predictors heavily influenced by precipitation), these hydrological drivers of chlorophyll-a are suppressed under SSP585 relative to SSP126. Although other variables, such as temperature, continue to increase across all scenarios and contribute positively to chlorophyll-a, their combined effect does not fully offset the stronger reduction in precipitation-driven inputs under SSP585, leading to the comparatively steeper chlorophyll-a increase under SSP126. It is worth noting that this pattern diverges from historical trends, where precipitation increased. Because our analysis relies on monthly climate data, short-term extremes that can trigger bloom initiation are not fully resolved, potentially underestimating the impacts of SSP585 on future HAB dynamics. Furthermore, our projections assume stationarity for unavailable water quality inputs (e.g.,

nutrients), which may compound this underestimation by overlooking climate-driven shifts in nutrient cycling. These limitations underscore the need for future studies to incorporate higher-resolution climate datasets, dynamically modeled nutrient inputs, and refined regional projections to better capture climate-driven shifts in hydrology and water quality.

The discrepancies between GCM projections and observed historical data present significant concerns regarding the reliability of these models in simulating future environmental conditions (Topál et al., 2022). Downscaling studies, particularly in regions like Southeast Florida, frequently encounter similar mismatches, such as historically increasing precipitation contrasted with projected declines (Rahimi et al., 2024; Raghavendra et al., 2019; Abiy et al., 2019). This divergence may stem from the GCMs' inherent limitations due to the complex interplay of multiple factors in spatiotemporal modeling, which they may not capture accurately (Hens et al., 2019). Alternatively, the coarse spatial low-resolution of GCMs may fail to represent localized climatic conditions adequately (Lee et al., 2014). Such issues highlight the need for improved regional climate models (RCMs) to capture nuanced local impacts (Rummukainen, 2016).

Overall, our findings suggest that while bias-corrected GCMs provide a robust framework for projecting climate scenarios, their ecological applications should be treated with caution, especially for phenomena like HABs. Enhancing model accuracy through regional calibration and incorporating a broader spectrum of climatic and non-climatic factors will improve the predictive reliability of future HABs (Ralston and Moore, 2020).

4.4. Strategic prediction: implications of projected HABs in North Biscayne Bay

In our analysis of future climate scenarios, we identified a consistent upward trend in chlorophyll-a concentrations in North Biscayne Bay under different SSPs (Fig. 6). Fig. 6a clearly illustrates these trends, with projections extending to 2100 showing increasing chlorophyll-a levels under all scenarios. Notably, SSP126 exhibits the most significant increases. While this upward trend aligns with the historical increases observed up to 2020, the increase in magnitude in the projected chlorophyll-a is less pronounced than in the historical records. This difference can be attributed, in part, to the inadequate capability of the models to predict extreme values (Fig. 3a). Besides, the potential for decreased precipitation across future scenarios, particularly with SSP585 showing the fastest decline and SSP126 the slowest fall (Fig. 4k), may contribute to these lower-than-realistic predictive increases in projected chlorophyll-a levels based on our feature importance analysis.

To further investigate the potential impacts of increasing trends, we also analyzed the frequency of extreme chlorophyll-a events under future scenarios, given that some studies suggest that while general trends might show a decline, extreme values could still rise (Raghavendra et al., 2019; Rahmstorf and Coumou, 2011). For this study, we defined an "extreme threshold" as 6.89 µg/L, calculated as the upper quartile (Q3) of observed chlorophyll-a concentrations plus 1.5 times the interquartile range (IQR), representing a statistically outlier level (Mishra et al., 2019). Table 4 summarizes this analysis, illustrating the frequency of predicted concentrations exceeding this extreme threshold. The analysis revealed varying frequencies of exceedances across models and scenarios, with the highest frequency observed under the SSP126 scenario and lower exceedances under SSP585.

The difference in frequency of exceedances and trends across scenarios suggests the relationship between climate change and HABs is nonlinear and sometimes competing (Gobler, 2020; Wells et al., 2020). The steeper increase in chlorophyll-a under SSP126 vs SSP585 may be due to the interaction of temperature, precipitation and nutrient delivery. Moderate warming under SSP126 may keep thermal conditions closer to the physiological optima for many bloom-forming phytoplankton species and enhance growth (Griffith and Gobler, 2020; Ralston and Moore, 2020). In contrast, the more extreme warming and precipitation decline under SSP585 may exacerbate stratification and reduce freshwater and nutrient inflows and limit nutrient availability and suppress bloom intensity despite higher temperatures (Li et al., 2020). This shows that extreme climate scenarios do not always translate to proportionally more bloom risk, and hydrological drivers are integrative controls on HAB dynamics in this system. Under future climate change, these nonlinear interactions will play out differently across regions, resulting in spatial or temporal expansion of HABs in some areas, such as the North Atlantic and North Pacific oceans (Gobler et al., 2017), East China Sea (Zhou et al., 2022), and northern Gulf of Mexico (Anderson et al., 2021), while being less evident or even suppressed in others. This highlights the need for local-scale HAB models and regional climate projections to capture system-specific responses (Dai et al., 2023; Ralston and Moore, 2020).

The seasonal analysis highlights important implications for HAB management in North Biscayne Bay under future climate scenarios (Fig. 6b). The consistently higher chlorophyll-a concentrations projected for the wet season than the dry season suggest that nutrient runoff and environmental factors contributing to HAB formation may become more pronounced during increased precipitation and temperature (Fig. 6b). This aligns with the feature importance analysis in Fig. 3b and findings from our previous studies (Yan et al., 2024a) and similar coastal regions (Xiao et al., 2019; Heil and Muni-Morgan, 2021; Griffith et al., 2019; Ryan et al., 2017). This seasonal variability amplifies over the long term, with both wet and dry seasons showing rising baselines, indicating sustained elevation of HAB risk (Wells et al., 2020). The results of Fig. 6b underscore the potential for heightened HAB frequency and intensity in the coming decades. These findings emphasize the need for seasonally tailored mitigation strategies, particularly targeting the wet season when HABs are likely to pose the region's ecological and public health risks (Gobler, 2020).

To our knowledge, very few studies have applied machine learning models in combination with bias-corrected climate projections to forecast chlorophyll-a or other ecological water quality indicators under future climate scenarios. While machine learning has been used for chlorophyll-a estimation recently (Kim et al., 2024; Wu et al., 2025) and bias-correction of climate variables is common in hydrological impact studies (Pierce et al., 2015; Teutschbein and Seibert, 2012), the integration of these approaches for projecting HAB-related indicators under future SSP forcing remains rare (Meema et al., 2025). This underscores the novelty of our framework, which links regionally adjusted, bias-corrected climate projections with interpretable machine learning models to simulate and predict future HAB scenarios under varying climate conditions. Although developed with site-specific data from BB02 in North Biscayne Bay,

Table 4

Frequency of predicted chlorophyll-a concentrations exceeding the extreme threshold by GCM Model and SSP Scenario, 2021–2100.

Scenario	BCC-CSM2-MR	CNRM-ESM2-1	MPI-ESM1-2-LR	MRI-ESM2-0	Ensemble mean
SSP126	57	82	52	36	48
SSP245	39	63	65	32	46
SSP585	24	54	49	17	29

the framework is generalizable to other estuarine systems, especially by combining explainable machine learning (SHAP) with non-autoregressive inputs, which decouples predictions from site-specific dependencies. Future applications can use similar hydro-meteorological and land-based predictors as long as region-specific calibration is done using local water quality data. Beyond Biscayne Bay, this modular framework can be adapted to regions with different data and management contexts and can be plugged into larger coastal monitoring programs. Coupling with real-time observational networks—such as buoy-based sensors and high-resolution, high-frequency remote sensing platforms—can detect bloom-favorable conditions early and inform proactive management (Vianna et al., 2023; Wells et al., 2020). The projected differences in overall trends and the frequency of extreme events across all scenarios offer critical insights for resource management and the development of effective mitigation strategies. Policymakers and environmental managers can use these findings to inform long-term strategic planning, ensuring proactive and adaptive responses to the impacts of HABs on ecological health and water quality. This approach is key to addressing the challenges posed by climate change and HABs.

5. Conclusion

Our study demonstrates that machine learning, when combined with bias-corrected climate projections, can robustly predict HABs in North Biscayne Bay. The SVM model achieved the highest accuracy, identifying precipitation, discharge, and temperature as dominant drivers of chlorophyll-a dynamics. Projections indicate increasing chlorophyll-a across all climate scenarios, with particularly elevated risks during the wet season and over long-term horizons, underscoring the importance of targeted monitoring and early-warning strategies between June and October. This work contributes a novel, climate-informed modeling framework that excludes autoregressive dependencies, integrates explainable machine learning, and provides transparent long-term forecasts. By doing so, it offers decision-relevant insights that can support adaptive watershed management, public health protection, and ecosystem resilience in urban estuaries. We acknowledge some limitations: the assumption of stationary water quality inputs, the omission of fine-scale precipitation extremes, and the lack of biological interactions or salinity dynamics. While these were necessary to ensure consistency across scenarios, they may underestimate future HAB risks under climate and land use change. Future efforts should incorporate dynamic nutrient models, regional climate projections, and coupled process-based frameworks to refine predictions. Overall, this study advances HAB prediction by bridging climate science with water quality management, delivering practical and reproducible tools that help safeguard ecosystems and communities against emerging climate-driven risks.

CRedit authorship contribution statement

Zhengxiao Yan: Writing – original draft, Visualization, Validation, Methodology, Formal analysis. **Nasrin Alamdari:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests. Nasrin Alamdari reports financial support was provided by Florida State University. Nasrin Alamdari reports a relationship with Florida State University that includes: employment and funding grants.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ejrh.2025.102865](https://doi.org/10.1016/j.ejrh.2025.102865).

Data availability

Data will be made available on request.

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